EXPLAINABLE VISUAL COMMONSENSE REASONING FOR SOCIAL INTERACTIONS

PHASE I REPORT

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BONAFIDE CERTIFICATE

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Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

The activity, relationship and intention behind a social interaction in an image are rarely visible directly; they must be inferred from visual cues together with everyday commonsense knowledge. This project builds an explainable Visual Commonsense Reasoning system that predicts these three aspects and supports every prediction with evidence that a user can inspect. In Phase I we built the prediction side of the system. OpenCV first cleans up and standardises the input image. A small vision-language model, SmolVLM, then describes who is in the scene, what they are doing and where they are. We look up the main concepts from that description in ConceptNet, and a log-linear fusion step combines both sources to give the three predictions. Each prediction comes with a confidence score, a ranked list of other likely labels and a short written explanation. We also fine-tuned SmolVLM on the VCR dataset using LoRA. Of the 20 training epochs, the checkpoint from epoch 11 worked best, and it lifted Q→AR accuracy on held-out VCR questions from 15.3% to 32.7%. The pipeline is exposed through a FastAPI–React interface and passes all 23 automated tests. Phase II will verify the explanations through visual attribution, cross-modal consistency and faithfulness evaluation, and present them in an explainability dashboard.

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CHAPTER 1

INTRODUCTION

Visual understanding has progressed from object recognition to higher-level reasoning about actions, relationships, context and human intention. Social interactions are particularly challenging because their interpretation depends on several people at once, on their posture and gestures, on the objects they use, on their spatial arrangement and on the wider context. Visual Commonsense Reasoning (VCR) extends conventional recognition by requiring a model to infer information that is not explicitly visible in an image [1].

This report presents the first phase of a system that performs such reasoning for social interactions and exposes the evidence behind each of its conclusions. OpenCV is used to bring every input into the same format. SmolVLM, a small vision-language model, then picks out the people in the scene along with their actions, gestures and surroundings. For each of these concepts we pull related everyday knowledge from ConceptNet, and a reasoning module weighs all of this evidence together to decide what the activity is, how the people are related and what they are trying to do. Each prediction is returned with a confidence value, ranked alternatives, supporting evidence and a textual explanation. SmolVLM is also fine-tuned on the VCR dataset with LoRA so that it reasons more reliably about people.

1.1 MOTIVATION

A photograph of two people shaking hands may show a business deal, a greeting between friends, a sporting gesture or a reconciliation after an argument. The pixels are much the same in each case; what differs is the context and the commonsense knowledge a person brings to the scene. Humans make these inferences effortlessly, but automated systems that make them are increasingly used in settings where the inference matters: assistive tools that describe scenes to visually impaired users, social robots that must decide how to approach people, and tools that summarise or moderate visual media.

Two problems limit the use of current models in such settings. The first is **opacity**. High-accuracy VCR models select an answer from a set of options but do not show which visual cues or which pieces of world knowledge led to the choice. Large generative vision-language models do produce explanations, but these explanations are not guaranteed to reflect the evidence the model actually used, and they can describe details that are not present in the image. **Cost** is the other problem. The strongest multimodal models have tens of billions of parameters and require data-centre hardware, which rules them out for on-device, privacy-preserving or low-cost deployment.

This project is motivated by the need for a system that is both *lightweight* and *inspectable*: it should run on modest hardware, and every prediction about a social interaction should come with structured evidence – the visual concepts detected, the commonsense facts used, the probabilities from each source and the alternatives that were considered – that a user can check.

1.2 BACKGROUND

1.2.1 Visual Commonsense Reasoning

The VCR task, introduced by Zellers et al. [1], poses a multiple-choice question about an image taken from a movie scene. The people and objects in the image are referred to by tags such as *person1* and *person2*. A model must solve two sub-tasks: Q→A, choosing the correct answer from four options, and QA→R, choosing the rationale that justifies the correct answer from four options. The joint metric Q→AR counts a question as correct only when both the answer and the rationale are correct. With four options each, chance accuracy is 25% for Q→A and QA→R and 6.25% for Q→AR.

1.2.2 Vision-Language Models

A vision-language model (VLM) couples an image encoder with a language model so that it can describe images and answer questions about them. Modern VLMs encode the image into a sequence of visual tokens, project them into the embedding space of a pre-trained language model, and generate text autoregressively [10, 11]. SmolVLM [12] is a family of compact VLMs designed for efficient inference. We use the `SmolVLM-Instruct` variant, which has roughly 2.2 billion parameters. It pairs a SigLIP vision encoder [13] with a small language model, and a connector between the two shrinks the number of visual tokens the language model has to read.

1.2.3 Commonsense Knowledge Graphs

ConceptNet [14] is a multilingual knowledge graph of everyday knowledge. Its nodes are words and phrases (for example *concert*, *sing*, *stage*) and its edges are labelled relations such as *RelatedTo*, *IsA*, *UsedFor*, *AtLocation*, *CapableOf*, *Antonym* and *DistinctFrom*, each with a confidence weight. Because each edge is a readable statement ("a performer is typically found at a stage"), ConceptNet is well suited to producing evidence that a human can inspect.

1.2.4 Parameter-Efficient Fine-Tuning

Fine-tuning all the weights of a multi-billion-parameter model is expensive. Low-Rank Adaptation (LoRA) [19] freezes the pre-trained weight matrix $W_0 \in \mathbb{R}^{d\times k}$ and learns a low-rank update:

$$h = W_0 x + \frac{\alpha}{r} BAx, \qquad B \in \mathbb{R}^{d\times r},\ A \in \mathbb{R}^{r\times k},\ r \ll \min(d,k) \tag{1.1}$$

where r is the rank and α a scaling factor. Only A and B are trained, so the number of trainable parameters and the optimiser memory are a small fraction of full fine-tuning, and after training BA can be merged into W_0 so that inference is no slower than with the base model.

1.2.5 Explainability and Faithfulness

An explanation is *plausible* if it convinces a human and *faithful* if it reflects the reasoning the model actually performed. Visual explanation methods such as

Grad-CAM [21] highlight the image regions that most influence a prediction. Their faithfulness can be measured with the Insertion–Deletion metric [22], which tracks how the model's confidence changes as the highlighted pixels are added or removed, and with the Pointing Game [23], which checks whether the most salient point falls inside the relevant object. Comparing what a textual explanation says with what a visual explanation highlights gives a measure of cross-modal consistency [25, 26]. Phase II of this project is built around these methods.

1.3 PROBLEM DEFINITION

Correctly detecting the people and objects in a scene does not mean the interaction has been understood. Two people and their gestures can be recognised perfectly, and the scene can still fit several different activities. Choosing between them, or working out how the people are related and why they are acting as they are, needs commonsense knowledge on top of what is visible. The problem addressed in this project is therefore:

> *Given a static image, video frame, or live-camera frame containing zero or more people, develop a lightweight system that identifies the visual concepts in the scene, enriches them with commonsense knowledge, and predicts the activity, relationship, and intention of a social interaction with interpretable evidence.*

1.3.1 Inputs

The main input is a single image, given as a file path, a URL, a NumPy array or a PIL image. The system can also take frames sampled from a video file or captured from a live camera. The user can optionally ask a question about the scene in their own words, and can add a set of multiple-choice answers for that question.

1.3.2 Outputs

The system's main output is a structured prediction covering three things: the activity, the relationship and the intention. For each one it gives a label, a confidence value and a ranked list of alternative labels. Every alternative also keeps the three scores it was built from: its VLM probability, its commonsense probability and its draft-match value. Along with the prediction, the system returns the evidence that supports it and a generated explanation in plain text. The intermediate results are returned as well: the visual concepts and draft reasoning produced by the VLM, the ConceptNet facts and label-support scores, and the processing time of each module.

1.3.3 Constraints

The system is built around lightweight multimodal models so that it can run on modest hardware. The ConceptNet integration uses a local SQLite database for reproducible offline operation and can fall back to other configured backends. The set of labels the system chooses from is not hard-coded. It is defined in `configs/taxonomy.yaml` and can be edited there.

1.3.4 Scope of Phase I

The full framework is organised as the following sequence of stages:

Input → Preprocessing → VLM → Commonsense → Reasoning → Explanation → Consistency → Evaluation → Dashboard

Phase I, reported here, covers the first four modules – Input and Preprocessing, the Small Vision-Language Model, ConceptNet Commonsense Enrichment and Social Interaction Reasoning – together with the offline fine-tuning of SmolVLM on VCR. Its deliverable is a working pipeline that takes visual input and produces a social-interaction prediction with a draft explanation. Phase II adds the textual and visual explanation paths, cross-modal consistency analysis, explainability evaluation and the final explainability dashboard.

1.4 ORGANIZATION OF THE REPORT

The rest of this report is organised into six chapters, each covering one main part of the work. Chapter 1 explains why the project was taken up, gives the necessary background and states the problem. Chapter 2 surveys the literature on visual commonsense reasoning, vision-language models, knowledge-augmented reasoning, social relation recognition and explainability, identifies the gaps this project addresses and states the research objectives.

Chapter 3 gives the methodology in detail. It covers the overall architecture, each Phase I module with its algorithm, how SmolVLM was fine-tuned with LoRA,

and how the modules fit together. Chapter 4 covers the datasets, the hardware and software we used, and the experiments we ran, followed by an analysis of the results.

Chapter 5 looks at the social impact of the system and how sustainable it is. Chapter 6 sums up the report and sets out what is planned for Phase II. The appendices list the configuration parameters and the API endpoints of the implementation.

CHAPTER 2

LITERATURE SURVEY

This chapter reviews work in five areas that bear on the problem: models and datasets for visual commonsense reasoning, pre-trained and instruction-tuned vision-language models, reasoning with external knowledge, recognition of social relations, and the explanation and evaluation of multimodal models.

2.1 PAPERS RELATED TO THE PROBLEM IDENTIFIED

2.1.1 Recognition to Cognition Network (R2C)

Zellers et al. [1] introduced the VCR dataset together with a baseline model, the Recognition-to-Cognition network (R2C). R2C first links the words in the question and in each candidate response to the object regions detected in the image. It then interprets each response in light of the question and the image, and finally reasons over this combined representation to give every option a score. To build the dataset, the authors used a method they call *Adversarial Matching*: the correct answer to one question is reused as a wrong option for another, so the distractors read naturally and stay on topic. Their results showed that even the best model was far behind humans, and VCR has since served as a benchmark for understanding at the level of cognition rather than recognition.

2.1.2 Tags and Attributes for VCR (TAB-VCR)

TAB-VCR, proposed by Lin et al. [2], is a deliberately simpler baseline. It attaches attributes to each detected object. When the text mentions an object that was never tagged in the image, it adds a new tag for that object. Better grounding alone was enough to beat R2C with fewer parameters, which suggests that tying the text to the right image regions matters more than building a more complex model.

2.1.3 Heterogeneous Graph Learning (HGL)

Heterogeneous Graph Learning, from Yu et al. [3], builds two graphs: one linking the visual content to the answers and one linking the question to the answers. A contextual voting module on top of these lets the model use visual context from distant parts of the image. The relations between visual and language elements are therefore modelled explicitly, but all of this reasoning stays inside the network and is never shown to the user.

2.1.4 VisualCOMET

VisualCOMET, introduced by Park et al. [4], is a dataset of over a million written commonsense inferences about still images. For each image it records what probably happened just before, what is likely to happen next and what the people in it intend. The task is to generate these inferences, not to pick from a list of options. That is much closer to the kind of explanation our system aims to give.

2.1.5 Pre-trained Vision-Language Transformers

ViLBERT [5] split BERT into two streams, one for the image and one for the text, which exchange information through co-attention layers. It was pre-trained on image–caption pairs and then transferred to downstream tasks, VCR among them. VL-BERT [6] used a single-stream transformer over both modalities, and UNITER [7] learned a universal image–text representation from several large datasets with word–region alignment objectives, reaching state-of-the-art VCR accuracy at the time. MERLOT [8] learned multimodal script knowledge from millions of YouTube videos and their transcripts, and transferred well to VCR. These models raised VCR accuracy considerably, but they are discriminative classifiers: they score given options and provide no explanation of their own beyond attention weights, which are not reliable indicators of the evidence used.

2.1.6 Generative Vision-Language Models

CLIP [9] trained image and text encoders contrastively on 400 million image–text pairs, enabling zero-shot recognition from natural-language labels. BLIP-2 [10] connected a frozen image encoder to a frozen large language model through a lightweight Querying Transformer, and LLaVA [11] connected a CLIP encoder to an instruction-tuned language model and trained it on machine-generated visual instruction data. These generative VLMs answer open questions and produce fluent free-form explanations, but the largest of them are expensive to run and their explanations can include details that are not present in the image.

2.1.7 SmolVLM: Compact Vision-Language Models

SmolVLM [12] goes in the opposite direction. Its models are kept small and compress the visual tokens heavily, so they can run on a device with limited memory. Its vision encoder is based on SigLIP [13]. SmolVLM makes a VLM practical on a laptop, which is why it is used in this project, but on its own it is not adapted to commonsense reasoning about people.

2.1.8 Knowledge-Augmented Visual Reasoning

ConceptNet 5.5 [14] is a large graph of general knowledge, and each of its edges can be read as a plain statement. OK-VQA [15] is a VQA benchmark whose questions cannot be answered from the image alone and require external knowledge. KRISP [16] combined the implicit knowledge of a pre-trained transformer with symbolic knowledge from graphs, including ConceptNet, through a graph network, and showed that both kinds of knowledge contribute. The catch is that these systems usually turn the retrieved knowledge into hidden feature vectors. The user has no way to see which facts actually shaped the answer.

2.1.9 Social Relation Recognition

Sun et al. [17] annotated social relations in the PIPA photo-album dataset following a domain-based theory from social psychology and predicted relations from attributes of the people such as age, gender, clothing and proximity. Li et

al. [18] introduced the PISC dataset and a Dual-Glance model that first looks at the pair of people and then attends to contextual objects in the scene. These works show that context and person attributes determine the relationship, but they treat the relationship as a single classification target and do not reason jointly about the activity and the intention.

2.1.10 Parameter-Efficient Adaptation

LoRA [19] adapts large models by training low-rank updates to frozen weight matrices, and QLoRA [20] combines LoRA with 4-bit quantisation of the frozen weights so that large models can be fine-tuned on a single GPU. These methods make it feasible to adapt a VLM to VCR on a single laboratory GPU.

2.1.11 Explanation Methods and Faithfulness Evaluation

Grad-CAM [21] produces class-specific saliency maps from the gradients flowing into the last convolutional or attention layer. RISE [22] explains black-box models by random masking and introduced the Insertion and Deletion metrics for evaluating saliency maps. The Pointing Game [23] checks whether the maximum of a saliency map lies on the annotated object. e-ViL [24] provides a benchmark for natural-language explanations in vision-language tasks. Other recent studies ask whether the textual explanations a VLM gives are consistent with each other and faithful to how it actually used the image and the text [26], and whether multimodal models answer the same question consistently across modalities [25].

Table 2.1 lists the main works covered above, together with where each falls short for the problem set out in Chapter 1.

TABLE 2.1: Summary of Related Work

Work	**Approach**	**Limitation for this problem**
R2C [1]	Grounded attention over detections; VCR benchmark	Selects options; no evidence shown to the user
TAB-VCR [2]	Attributes and new tags for better grounding	Same discriminative setting; no external knowledge
HGL [3]	Heterogeneous vision/question–answer graphs	Graph reasoning is internal and not readable
ViLBERT, VL-BERT, UNITER [5–7]	Large-scale vision–language pre-training	High accuracy but opaque; heavy fine-tuning
MERLOT [8]	Script knowledge from video	Very large pre-training cost
BLIP-2, LLaVA [10, 11]	Generative VLMs with LLM backbones	Fluent but unverified explanations; large models
SmolVLM [12]	Compact efficient VLM	Not adapted to social commonsense reasoning
KRISP [16]	Implicit + symbolic (ConceptNet) knowledge	Knowledge hidden in features; not social-specific
Dual-Glance, PIPA [17, 18]	Social relation classification with context	Relationship only; no activity or intention
Grad-CAM, RISE, Pointing Game [21–23]	Saliency maps and their evaluation	Not linked to textual explanations
e-ViL [24]	Benchmark for natural-language explanations	Evaluates text alone, not cross-modal agreement

2.2 RESEARCH GAPS

From this survey we identified six gaps.

- **G1. Accuracy without explanation:** The best-performing VCR models pick an answer and a rationale from the options they are given, but they never show which visual cues or which pieces of knowledge led to that choice.

- **G2. Explanations without verification:** Generative VLMs write fluent explanations, but nothing checks these against the image, so they sometimes mention things that are not there at all.

- **G3. Hidden knowledge:** Knowledge-augmented models absorb the facts they retrieve into hidden representations. As a result, the user cannot tell which facts mattered for a given prediction.

- **G4. Single-target social understanding:** Work on social relation recognition predicts the relationship as a label on its own. The activity, relationship and intention are not predicted together, so nothing keeps them consistent with each other.

- **G5. Separate evaluation of explanations:** Visual saliency maps and textual explanations are usually evaluated separately, and whether they agree is rarely measured.

- **G6. Computational cost:** Most high-performing systems depend on large models and data-centre hardware, which makes them impractical for lightweight or on-device use.

2.3 RESEARCH OBJECTIVES

Table 2.2 lists the objectives we set to close these gaps. Objectives 1 to 10 belong to Phase I and are covered in this report. Objectives 11 to 13 are left for Phase II.

TABLE 2.2: Identified Research Gaps and Proposed Research Objectives

Category	Identified Gaps	Proposed Research Objectives
Lightweight Visual Understanding	Large models and data-centre hardware required (G6).	**1.** Build an OpenCV-based input and preprocessing pipeline that reliably handles still images, video frames and live-camera input. **2.** Use the lightweight SmolVLM both to understand the image and to reason about it in language. **3.** Fine-tune SmolVLM on VCR with LoRA, and pick the best checkpoint using held-out questions.
Inspectable Evidence	Opaque predictions and hidden knowledge (G1, G3).	**4.** Extract the people, objects, actions, gestures, setting and spatial layout of a scene as structured concepts. **5.** Add ConceptNet knowledge to these concepts and show it as evidence paths a person can read. **6.** Build a reasoning module that combines the visual concepts, the commonsense knowledge and the VLM's draft answers.
Joint Social Understanding	Relationship predicted in isolation (G4).	**7.** Predict activity, relationship and intention in sequence so that they are mutually consistent. **8.** Provide confidence values, ranked alternatives and supporting evidence for each prediction.
Integration and Validation	Research prototypes rarely usable end to end.	**9.** Integrate the modules into a reusable Python pipeline and a FastAPI–React demonstration interface. **10.** Validate the modules with automated unit and end-to-end tests.
Verified Explanations (Phase II)	Unverified explanations; separate evaluation of modalities (G2, G5).	**11.** Generate independent textual and visual (Grad-CAM) explanations for each prediction. **12.** Measure cross-modal consistency and evaluate faithfulness with the Insertion–Deletion metric and the Pointing Game. **13.** Present predictions, explanations and scores in an explainability dashboard.

CHAPTER 3

PROPOSED METHODOLOGY

This chapter describes the architecture of the proposed system and the design of each Phase I module. The Phase I pipeline is

Input → OpenCV Preprocessing → SmolVLM → ConceptNet → Social Reasoning

with an offline training phase in which SmolVLM is fine-tuned on VCR.

3.1 ARCHITECTURE DESIGN

The overall architecture of the proposed system, covering both phases, is shown in Figure 3.1. Visual input is preprocessed with OpenCV and passed to a small vision-language model, which can be replaced by a version fine-tuned offline on the VCR dataset. The output of the model is enriched with commonsense knowledge, and the social interaction reasoning module fuses the two into a prediction.

In Phase II, this prediction is explained along two independent paths, one textual and one visual. The consistency between the two explanations is measured, their faithfulness is evaluated with the Insertion–Deletion metric and the Pointing Game, and the results are presented in an explainability dashboard.

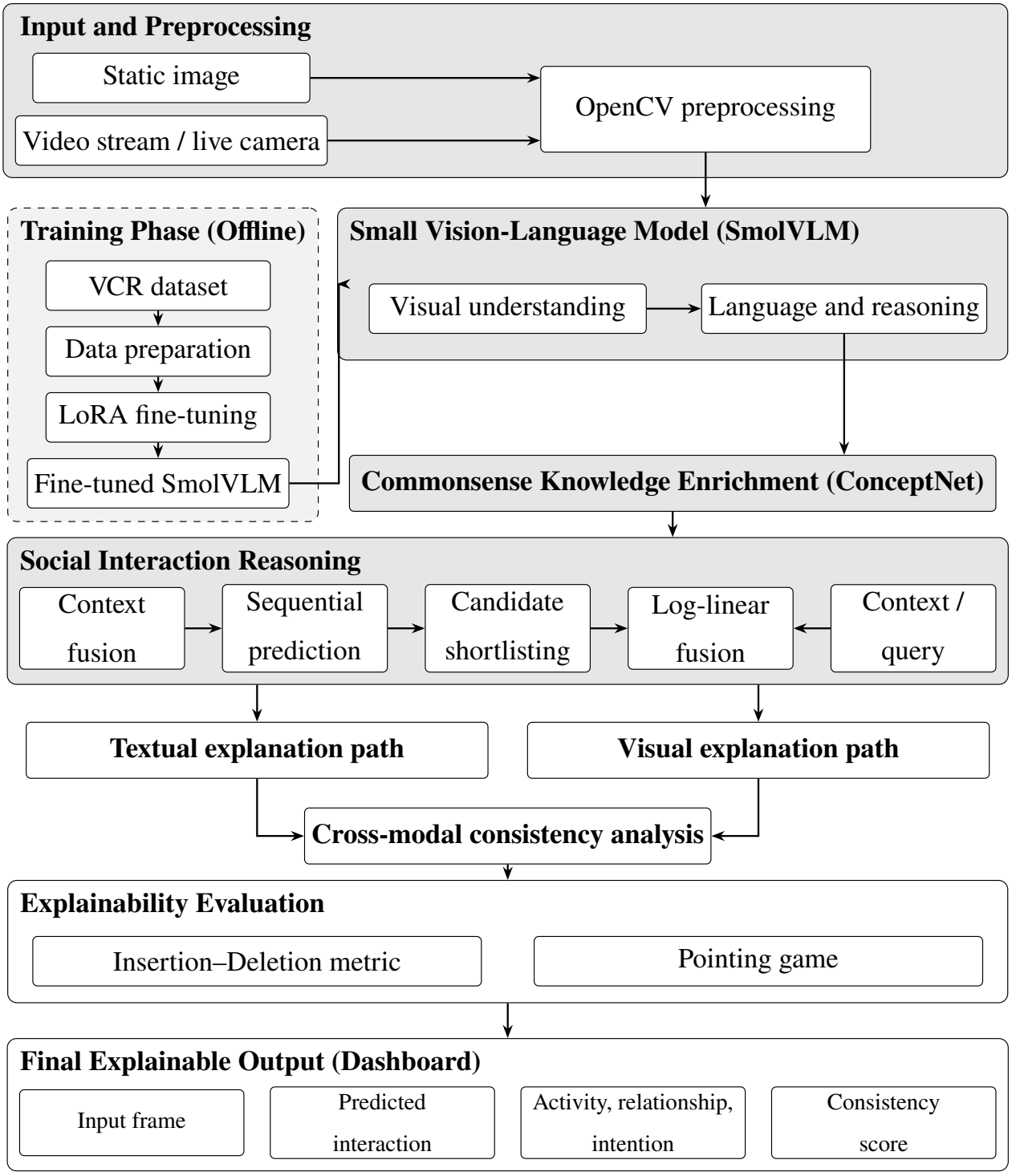


FIGURE 3.1: Architecture of the Proposed System (Phase I modules shaded, Phase II modules unshaded)

3.2 PHASE I ARCHITECTURE

Figure 3.2 shows the part of the architecture built in Phase I. The modules do not pass raw text from one stage to the next. They hand structured Python data objects to each other. The main implementation files are listed in Table 3.1.

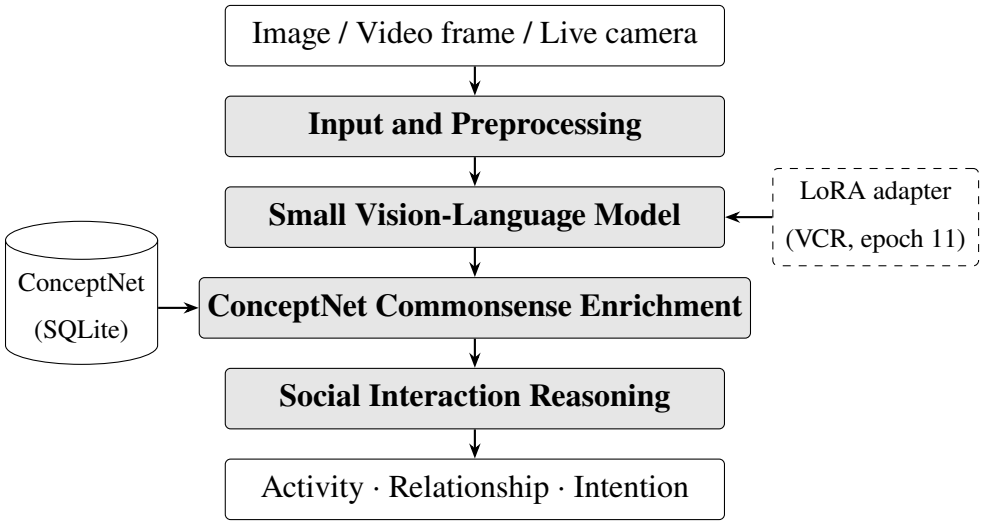


FIGURE 3.2: Phase I Architecture

TABLE 3.1: Phase I Module-to-Code Mapping

Module	**Implementation**
Input and Preprocessing	`svcr/preprocessing.py`, `Preprocessor`, `FrameStream`
Small Vision-Language Model	`svcr/vlm.py`, `SmolVLMEngine`
ConceptNet Commonsense Enrichment	`svcr/conceptnet/client.py`, `enrichment.py`, `relations.py`
Social Interaction Reasoning	`svcr/reasoning.py`, `SocialInteractionReasoner`
Fine-Tuning (offline)	`finetune/train_vcr_lora.py`, `configs/finetuned.yaml`
Pipeline Integration	`svcr/pipeline.py`, `Phase1Pipeline`
Data Structures	`svcr/schemas.py`
Configuration	`configs/default.yaml`, `configs/taxonomy.yaml`

3.3 DATA FLOW

For a single image, data flows through the pipeline as follows. The image is acquired and its quality is assessed; the necessary preprocessing is applied, and the frame is converted into the RGB representation expected by the model together with a normalised tensor. SmolVLM then looks at the frame and picks out the people, objects, actions, gestures, setting and the spatial layout of the scene. From this it writes a first draft of the activity, relationship and intention, with a rationale. If the user has asked a question, it answers that too.

Salient concepts are next extracted from these outputs and looked up in ConceptNet. The useful commonsense relations are retained and used to calculate the support for each candidate label in the taxonomy. Finally, the VLM probabilities, the ConceptNet support and the draft-matching scores are fused to predict the activity, relationship and intention in sequence, and the structured result is returned with its evidence, alternatives, explanation and timings.

3.4 MODULES

The Phase I system consists of the following modules: Input and Preprocessing, the Small Vision-Language Model, Fine-Tuning on VCR (offline), ConceptNet Commonsense Enrichment, Social Interaction Reasoning, and the Phase I Integration that connects them.

3.4.1 Input and Preprocessing Module

The Input and Preprocessing module provides a common input representation for all downstream components. The code is in `svcr/preprocessing.py`. It relies mainly on OpenCV [27] and NumPy, and uses PIL only where needed.

3.4.1.1 Input Sources

The module accepts image file paths, HTTP and HTTPS image URLs, NumPy arrays and PIL images, as well as video files and live camera streams. Each is converted into a common BGR frame before further processing.

3.4.1.2 Quality Assessment

For every frame the brightness, contrast and blur are calculated; blur is estimated from the variance of the Laplacian. Frames that are dark or low in contrast are enhanced by applying CLAHE [28] to the luminance channel of the LAB colour space, followed by a mild gamma correction. Optional non-local-means denoising is available through the configuration.

3.4.1.3 Resize and Normalisation

The default resizing mode preserves the aspect ratio and limits the longest side to a size compatible with the model. The RGB image is retained as an 8-bit image for

the SmolVLM processor, and a normalised tensor is also generated:

$$x_{\text{norm}} = \frac{x/255 - \mu}{\sigma}, \qquad \mu = \sigma = (0.5, 0.5, 0.5) \tag{3.1}$$

The tensor has shape $1 \times 3 \times H \times W$ and is retained for the Phase II operations, Grad-CAM and Insertion–Deletion. The procedure is given in Algorithm 1.

Algorithm 1 Input Preprocessing

Input: Image source S (path, URL, array or PIL image); target longest side N; normalisation constants μ, σ

Output: Preprocessed frame F

$I \leftarrow \textsc{LoadAsBGR}(S)$; $(h, w) \leftarrow$ size of I
$g \leftarrow$ greyscale version of I
$b \leftarrow \text{mean}(g)$; $c \leftarrow \text{std}(g)$; $z \leftarrow \text{Var}(\nabla^2 g)$ ▷ brightness, contrast, blur
if $b < 60$ **or** $c < 25$ **then**
    $I \leftarrow$ CLAHE on the L channel of LAB, then gamma 0.8
end if
if denoising is enabled **then**
    $I \leftarrow \textsc{NonLocalMeansDenoise}(I)$
end if
$s \leftarrow N/\max(h, w)$
if $s < 1$ **then**
    $I \leftarrow \textsc{Resize}(I, s)$ ▷ aspect ratio preserved
end if
$R \leftarrow \text{BGRtoRGB}(I)$
$T \leftarrow (R/255 - \mu)/\sigma$, arranged as $1 \times 3 \times H \times W$
return $F = \big(R, T, (w, h), s, \{b, c, z, \text{blurry} = z < 60\}\big)$

3.4.1.4 Video and Live-Camera Handling

For video and live-camera input, `FrameStream` samples frames at a configurable interval. Blurry frames can be skipped, and frames that show the same scene as the last analysed frame are filtered out using the Bhattacharyya distance between HSV histograms, with a periodic refresh so that a scene is not suppressed indefinitely (Algorithm 2). For uploaded videos, the web interface selects up to seven key moments: the clip is divided into scene segments using colour histograms, and the sharpest frame from the central part of each segment is analysed.

Algorithm 2 Frame Sampling for Video and Live Camera

Input: Frame stream $\{(f_t, t)\}$; interval Δ; scene-change threshold τ

Output: Sequence of frames to analyse

$t_{\text{check}} \leftarrow -\infty$; $t_{\text{last}} \leftarrow -\infty$; $H_{\text{last}} \leftarrow \text{NULL}$
for all frames (f_t, t) in the stream **do**
  if $t - t_{\text{check}} < \Delta$ **or** f_t is blurry **then**
    continue
  end if
  $t_{\text{check}} \leftarrow t$; $H \leftarrow$ HSV histogram of f_t
  if $H_{\text{last}} \neq \text{NULL}$ **and** $t - t_{\text{last}} < 3\Delta$ **and** $d_{\text{Bhatt}}(H_{\text{last}}, H) < \tau$ **then**
    continue ▷ same scene as last analysed frame
  end if
  $t_{\text{last}} \leftarrow t$; $H_{\text{last}} \leftarrow H$
  output f_t
end for

3.4.2 Small Vision-Language Model Module

The VLM module is implemented by `SmolVLMEngine` in `svcr/vlm.py`. The default configuration uses `HuggingFaceTB/SmolVLM-Instruct`, loaded through Hugging Face Transformers [29]. The model and its processor are loaded only once, and the engine runs on CUDA, MPS or the CPU, depending on what the machine has. It exposes three operations. `generate()` produces text deterministically, `score_options()` estimates the probability of each option in a multiple-choice question, and `run()` carries out the whole visual-understanding and draft-reasoning stage.

3.4.2.1 Visual Understanding

The engine starts by writing a detailed description of the image that focuses on the people in it. It then asks a series of short follow-up questions: how many people there are, which objects matter, what actions are taking place, what facial expressions, gestures and body language are visible, what the setting is, and what each person looks like or what role they seem to play. The answers are parsed into clean lists of concepts and saved in a `VisualConcepts` object. A set of text utilities tidies the phrasing and also extracts spatial relationships from the description.

3.4.2.2 Language and Reasoning

In the second sub-stage, the engine writes free-form draft answers for the activity, the relationship and the intention. It also writes a short visual rationale for these

answers, and answers the user's question if there is one. Decoding is kept deterministic: sampling is turned off and a repetition penalty is added, so the same input always gives exactly the same output. Algorithm 3 gives the procedure; GENERATE denotes greedy decoding with a repetition penalty.

Algorithm 3 Scene Description and Draft Reasoning with SmolVLM

Input: Preprocessed frame F; optional question Q

Output: Visual concepts V; draft reasoning D

$cap \leftarrow \textsc{Generate}(F, \text{describe-the-people prompt})$

$\theta \leftarrow \textsc{ExtractTerms}(cap)$ ▷ nouns, verbs, adjectives, people

for all $k \in \{\text{count, objects, actions, gestures, setting, people}\}$ **do**

    $a_k \leftarrow \textsc{Generate}(F, \text{prompt}_k)$

end for

$n \leftarrow \textsc{ParseCount}(a_{\text{count}})$; if no number is found, $n \leftarrow \textsc{ParseCount}(cap)$

$V.objects \leftarrow \textsc{Split}(a_{\text{objects}}) \cup \theta.\text{nouns}$

$V.actions \leftarrow \textsc{Split}(a_{\text{actions}}) \cup \theta.\text{verbs}$

$V.gestures \leftarrow \textsc{Split}(a_{\text{gestures}})$; $V.people \leftarrow \textsc{Split}(a_{\text{people}}) \cup \theta.\text{people}$

$V.setting \leftarrow a_{\text{setting}}$; $V.spatial \leftarrow \textsc{SpatialRelations}(cap)$

if $n \geq 2$ **then**

    $D.activity \leftarrow \textsc{Generate}(F, \text{“what interaction is happening?”})$

    $D.relationship \leftarrow \textsc{Generate}(F, \text{“what is their relationship?”})$

else

    $D.activity \leftarrow \textsc{Generate}(F, \text{“what is the person doing?”})$

    $D.relationship \leftarrow$ “no relationship”

end if

$D.intention \leftarrow \textsc{Generate}(F, \text{“why are they } D.activity\text{?”})$

$D.rationale \leftarrow \textsc{Generate}(F, \text{“what visible evidence shows this?”})$

if $Q \neq \textsc{null}$ **then**

    $q' \leftarrow Q +$ “answer briefly and give the visual reason”

    $D.answer \leftarrow \textsc{Generate}(F, q')$

end if

return (V, D)

3.4.2.3 Multiple-Choice Scoring

For a multiple-choice question the engine reads the next-token logits for the option letters instead of generating text. The resulting option probabilities are averaged over several orderings of the options to reduce letter and position bias (Algorithm 4), and are later used by the reasoning module.

Algorithm 4 Debiased Multiple-Choice Scoring

Input: Frame F; question q; options $o_1, \ldots, o_n$; context c; number of orderings k

Output: Probabilities $p_1, \ldots, p_n$ with $\sum_i p_i = 1$

if $n = 1$ **then**
    return (1)
end if
$\Pi \leftarrow$ {identity order}; if $k \geq 2$, add the reversed order and $k - 2$ cyclic shifts
$p \leftarrow (0, \ldots, 0)$
for all orderings $\pi \in \Pi$ **do**
    $prompt \leftarrow c + q +$ one line "letter$_i$. $o_{\pi(i)}$" for each option
    append "Answer with the letter of the best option." to *prompt*
    $z \leftarrow$ next-token logits of SmolVLM for $(F, prompt)$
    for $i = 1$ **to** n **do**
        $\ell_i \leftarrow \max_{t \in \text{tokens(letter}_i)} z_t$
    end for
    $s \leftarrow \text{softmax}(\ell)$
    for $i = 1$ **to** n **do**
        $p_{\pi(i)} \leftarrow p_{\pi(i)} + s_i$ ▷ map back to the original option
    end for
end for
return $p/|\Pi|$

3.4.3 Fine-Tuning on VCR (Offline)

Used zero-shot, SmolVLM sometimes gives reasons that the image does not support. To adapt it to commonsense reasoning about people, the model is fine-tuned on the VCR dataset [1] with LoRA [19] (Equation 1.1). Low-rank adapters of rank 16 are added to the attention and feed-forward projections of the language model, while the SigLIP vision encoder and the connector stay frozen, so only a small fraction of the parameters is trained (about 18.1 million trainable parameters).

3.4.3.1 Data Preparation

The people and objects referred to in each VCR question are drawn on the image as labelled boxes (*person1*, *person2*, and so on), and the question text uses the same tags. Three kinds of training example are built from each question: multiple-choice question answering (Q→A), multiple-choice selection of the rationale for the correct answer (QA→R), and a free-form answer followed by its visual reason. The multiple-choice prompts are word for word the prompts used by the option scoring of Algorithm 4, so training directly sharpens the probabilities that the reasoning module fuses. The order of the options is shuffled so that no letter is favoured, and the loss is computed only on the answer tokens.

3.4.3.2 Training and Checkpoint Selection

The model is trained for 20 epochs. After every epoch the adapter is saved and scored on a selection set of VCR validation questions, and the epoch with the

highest Q→AR accuracy is kept; ties are broken by higher Q→A accuracy and then by the earlier epoch. The final accuracy is measured on a separate test set of held-out validation questions, so the choice of epoch does not bias the reported result (Algorithm 5). In the completed run, epoch 11 was selected. The selected adapter is merged into the base weights when the model is loaded through `configs/finetuned.yaml`, so inference with the fine-tuned model is as fast as with the base model.

3.4.3.3 Implementation

The procedure is implemented in `finetune/train_vcr_lora.py`. The script streams VCR images from the dataset's Parquet files, draws the labelled boxes, builds the three kinds of training example and caches the prepared data. It saves a checkpoint every 50 steps so that an interrupted run can be resumed, and records the training loss, the per-epoch selection accuracy and the final test accuracy. After the run, `install_adapter.py` copies the selected adapter into the project. The configuration is listed in Table 3.2.

TABLE 3.2: Fine-Tuning Configuration

Setting	**Value**
Base model	`HuggingFaceTB/SmolVLM-Instruct` (2.2B parameters)
Adapter	LoRA, rank 16, alpha 32, dropout 0.05
Adapted layers	Query, key, value, output, gate, up and down projections of the language model
Frozen components	SigLIP vision encoder and connector
Training data	Hugging Face VCR dataset (Q→A, QA→R and explanation examples)
Image resolution	Longest edge 768 pixels
Batch size	2, with gradient accumulation over 8 steps (effective batch of 16)
Optimiser	AdamW, learning rate 10^{-4}, cosine schedule with 5% warm-up
Training length	20 epochs of 75 steps (1,500 optimiser steps)
Checkpoint selection	Best Q→AR on selection questions: epoch 11
Test set	Held-out VCR validation questions

Algorithm 5 LoRA Fine-Tuning with Best-Epoch Selection

Input: Base model M; training examples X; selection set S_{sel}; test set S_{test}; number of epochs $E = 20$

Output: Selected adapter A^*; test accuracy before and after fine-tuning

add LoRA adapters to the language model of M; freeze all other weights
$r_{\text{base}} \leftarrow \textsc{Evaluate}(M \text{ without adapters}, S_{\text{test}})$
for $e = 1$ **to** E **do**
    for all mini-batches $B \subset X$ **do**
        update the adapters to minimise the cross-entropy of the answer tokens of B
    end for
    $A_e \leftarrow$ current adapters; $r_e \leftarrow \textsc{Evaluate}(M + A_e, S_{\text{sel}})$
end for
$e^* \leftarrow \arg\max_e \left(r_e.\text{Q}{\rightarrow}\text{AR},\ r_e.\text{Q}{\rightarrow}\text{A},\ -e\right)$ ▷ ties: higher Q→A, then earlier epoch
$A^* \leftarrow A_{e^*}$; $r_{\text{ft}} \leftarrow \textsc{Evaluate}(M + A^*, S_{\text{test}})$
return $A^*, r_{\text{base}}, r_{\text{ft}}$

3.4.4 ConceptNet Commonsense Enrichment Module

Visual observations alone may not be enough to distinguish semantically similar interactions. The enrichment module uses ConceptNet [14] to add commonsense knowledge to the visual concepts and produces evidence that is fused into the final prediction. It is divided into a client layer, which handles backend access, caching, term normalisation and phrase back-off, and an enrichment layer, which assigns salience, filters evidence and computes label support.

3.4.4.1 Key-Term Extraction

Candidate terms are collected from the actions, gestures, roles, objects and setting identified by the VLM, from its description and draft answers, and from the user's question. Each term is given a salience weight. The kinds of terms most relevant to an interaction get the highest weights, such as 1.0 for actions and 0.9 for gestures.

3.4.4.2 ConceptNet Backend

The module can use three backends. The first is a local SQLite copy of ConceptNet. The second is the public ConceptNet API, with responses cached on disk. The third is an offline mode in which only keyword matching against the taxonomy is available. Whenever the local database is present, it is used first, because it gives the same results every time. If a phrase such as "shaking hands" is not found, the module tries `shake_hands` next. If that also fails, it falls back to the head word.

3.4.4.3 Relevance Filtering

For each term the module retrieves its edges and filters them by relation type and weight. The highest-ranked facts are those that connect a term either to the anchor concepts of the social-interaction labels or to other concepts seen in the same image. Generic hub concepts, which link to almost everything in the graph, are left out because they do little to separate one label from another.

3.4.4.4 Label Support and Ambiguity Resolution

Next, the module works out how strongly the collected evidence supports each activity, relationship and intention label (Algorithm 7). A term contributes support if it is one of the label's anchor concepts, if a direct edge connects it to an anchor, if its two-hop neighbourhood overlaps that of the label, or if it matches one of the label's keywords, weighted by inverse document frequency. Negative relations such as *Antonym* and *DistinctFrom* reduce the support. The module therefore produces a numerical support score for every label together with readable evidence paths, and combines the support with the match to the VLM draft to produce a preliminary disambiguated label for each task (Algorithm 6). The taxonomy, including each label's anchors and keywords, is kept in `configs/taxonomy.yaml` so that it can be changed without modifying the code.

Algorithm 6 ConceptNet Commonsense Enrichment

Input: Visual concepts V; draft D; optional question Q; taxonomy $\mathcal{T}$

Output: Facts $\mathcal{F}$; support S; draft match M; evidence; disambiguated labels $\hat{L}$

$E \leftarrow$ terms from V, D and Q, each with a salience (action 1.0, gesture 0.9, role 0.9, object 0.7, setting 0.6, caption 0.5); keep the top 20
for all terms $e \in E$ **do**
  $(c_e, edges_e) \leftarrow \textsc{Resolve}(e)$ ▷ phrase → lemma → head word
  if c_e is shorter than e **then**
    $\text{salience}(e) \leftarrow 0.5 \cdot \text{salience}(e)$ ▷ backed-off sense is less reliable
  end if
  $P_e[v] \leftarrow w_{\text{rel}} \cdot (1 - e^{-\text{weight}})$ for each neighbour v that is not a generic hub
  $N_e \leftarrow$ neighbours linked by *Antonym*, *DistinctFrom* or *Not** relations
end for
$\mathcal{F} \leftarrow$ top 6 facts per term, ranked by relevance (×2 if the neighbour is a label anchor, ×1.5 if it is another observed concept); keep the top 40
for all tasks $\tau \in \{\text{activity, relationship, intention}\}$ **do**
  for all labels $L \in \mathcal{T}[\tau]$ **do**
    $S_\tau(L) \leftarrow \dfrac{\sum_e w_{e,\tau}\, \textsc{Score}(e, L)}{\sum_e w_{e,\tau}}$ ▷ Algorithm 7
    $M_\tau(L) \leftarrow \max\left(\text{lex}(D_\tau, L),\ 0.8 \max_{e \in \text{draft}} \textsc{Score}(e, L)\right)$
  end for
  $S_\tau \leftarrow S_\tau / \max\left(\max_L S_\tau(L),\ 0.3\right)$
  $\hat{L}_\tau \leftarrow \arg\max_L \left(S_\tau(L) + M_\tau(L)\right)$
end for
return $\mathcal{F}, S, M$, evidence paths, $\hat{L}$

Here $w_{e,\tau}$ is the salience of term e multiplied by the weight of its kind for task τ; for example, roles count most for the relationship task and actions most for the activity task.

Algorithm 7 Support of One Evidence Term for One Label

Input: Term e with concept c_e, neighbour profile P_e and negative neighbours N_e; label L with anchors A_L, lexicon X_L and neighbour profile P_L

Output: Support score s

if $c_e \in A_L$ **then**
  return 1.0 ▷ exact anchor
end if
$s \leftarrow 0$
for all anchors $a \in A_L$ **do**
  if $a \in P_e$ **then**
    $s \leftarrow \max(s,\ 0.9\, P_e[a])$ ▷ direct edge
  end if
end for
$s \leftarrow \max\left(s,\ 0.6 \cdot \min(1,\ 2.5 \cos(P_e, P_L))\right)$ ▷ two-hop overlap
$H \leftarrow \text{words}(e) \cap X_L$
if $H \neq \emptyset$ **then**
  $s \leftarrow \max\left(s,\ 0.6/\min_{w \in H} \text{df}(w)\right)$ ▷ keyword, IDF-weighted
end if
if $N_e \cap A_L \neq \emptyset$ **then**
  $s \leftarrow s - 0.5$ ▷ contradicting relation
end if
return s

3.4.5 Social Interaction Reasoning Module

The Social Interaction Reasoning module, `SocialInteractionReasoner`, is the fusion point of Phase I. It combines the visual concepts, the commonsense evidence, the VLM draft and the optional user question to predict the social interaction.

3.4.5.1 Context Fusion

The module builds a structured context from the scene description, the number of people and their descriptions, the actions, gestures, objects and setting identified by the VLM, a selection of the most relevant ConceptNet facts, and the user's question. This context accompanies the image in every prompt that the module sends to the VLM.

3.4.5.2 Sequential Prediction

The three tasks are inferred in the order activity, relationship, intention. The predicted activity is added to the context for relationship inference, and both the activity and the relationship are added to the context for intention inference, so that each later prediction is consistent with the earlier ones.

3.4.5.3 Candidate Shortlisting

ConceptNet support, VLM draft matching and lexical evidence are combined into a prior over the taxonomy labels. When the prior is informative, only the highest-ranked candidates are shortlisted; otherwise the complete label set is passed to the VLM.

3.4.5.4 Log-Linear Fusion

Let $p_{\mathrm{VLM}}(L)$ be the VLM probability for label L, $p_{\mathrm{CN}}(L)$ the normalised ConceptNet support, and $m_{\mathrm{draft}}(L)$ the match between the VLM draft and the

label. The module computes

$$\log P(L) \propto w_{\text{VLM}} \log p_{\text{VLM}}(L) + w_{\text{CN}} \log p_{\text{CN}}(L) + w_{\text{draft}}\, m_{\text{draft}}(L) \tag{3.2}$$

with default weights $w_{\text{VLM}} = 1.0$, $w_{\text{CN}} = 0.5$ and $w_{\text{draft}} = 1.0$. The resulting distribution gives the highest-scoring label with its confidence, the ranked alternatives with their component probabilities, and the evidence behind the prediction. A prediction is marked ambiguous when the probability margin between the top two labels is below 0.15. Algorithm 8 infers one task in this way.

Algorithm 8 Inference of One Task with Log-Linear Fusion

Input: Task τ; frame F; context c; visual concepts V; support S_τ; draft match M_τ; disambiguated label $\hat{L}_\tau$

Output: Prediction for τ: label, confidence, ambiguity flag, alternatives, evidence

$\mathcal{L} \leftarrow$ labels of τ (excluding `none_single` for relationship)
for all $L \in \mathcal{L}$ **do**
    $prior(L) \leftarrow S_\tau(L) + 1.5\, M_\tau(L) + 0.3\, \mathrm{lex}(\text{scene text}, L)$
end for
if fewer than 3 labels have $prior > 0.05$ **then**
    $C \leftarrow \mathcal{L}$ ▷ no reliable prior: VLM sees all labels
else
    $C \leftarrow$ the 8 labels with the highest prior, always including $\hat{L}_\tau$
end if
$p_{\mathrm{VLM}} \leftarrow \textsc{ScoreOptions}(F, \text{question}_\tau, \text{descriptions of } C, c)$ ▷ Algorithm 4
if $\max_{L \in C} S_\tau(L) < 10^{-3}$ **then**
    $p_{\mathrm{CN}} \leftarrow$ uniform over C
else
    $p_{\mathrm{CN}} \leftarrow \mathrm{softmax}(S_\tau / T), T = 0.2$
end if
for all $L \in C$ **do**
    $z(L) \leftarrow w_{\mathrm{VLM}} \log p_{\mathrm{VLM}}(L) + w_{\mathrm{CN}} \log p_{\mathrm{CN}}(L) + w_{\mathrm{draft}}\, M_\tau(L)$
end for
$P \leftarrow \mathrm{softmax}(z)$; sort C by P in descending order
$L^* \leftarrow C_1$; $ambiguous \leftarrow \big(P(C_1) - P(C_2) < 0.15\big)$
return $\big(L^*, P(L^*), ambiguous, C \text{ with } P, p_{\mathrm{VLM}}, p_{\mathrm{CN}}, M_\tau, \text{evidence for } L^*\big)$

3.4.5.5 Special Cases and Explanation

Scenes with zero or one detected person are handled explicitly. When only one person is present, the relationship is set to `none_single`; when no people are

detected, the system returns a no-interaction prediction instead of inferring a social relationship. Once the three tasks are inferred, an explanation is generated by a prompt that asks the VLM to relate the prediction to the visible people, gestures, facial expressions, objects and setting. Algorithm 9 applies these steps to the three tasks in sequence.

Algorithm 9 Social Interaction Reasoning

Input: Frame F; visual concepts V; draft D; commonsense result R; optional question Q and options O

Output: Social prediction: activity, relationship, intention, explanation, answer

$c \leftarrow \textsc{Fuse}(V, R, Q)$ ▷ caption, concepts, top ConceptNet facts, question
if $V.people = 0$ **then**
  return no-interaction prediction
end if
$A \leftarrow \textsc{InferTask}(\text{activity}, F, c)$ ▷ Algorithm 8
if $V.people = 1$ **then**
  $Rel \leftarrow$ `none_single` with confidence 1
else
  $Rel \leftarrow \textsc{InferTask}(\text{relationship}, F, c + \text{“The people are } A.\text{”})$
end if
$c' \leftarrow c +$ “The people are A. They are most likely Rel.”
$I \leftarrow \textsc{InferTask}(\text{intention}, F, c')$
$pr \leftarrow$ statement of (A, Rel, I) + top ConceptNet facts + “explain which visible evidence supports this”
$Ex \leftarrow \textsc{Generate}(F, pr)$
if $Q \neq \textsc{null}$ **and** $O \neq \textsc{null}$ **then**
  $p \leftarrow \textsc{ScoreOptions}(F, Q, O, c)$; $ans \leftarrow O_{\arg\max p}$
else if $Q \neq \textsc{null}$ **then**
  $ans \leftarrow D.answer$
end if
return (A, Rel, I, Ex, ans) with the supporting evidence

3.4.6 Phase I Integration

The four modules are integrated through the `Phase1Pipeline` class, which runs them in sequence (Algorithm 10). The modules exchange structured dataclasses rather than free text: `PreprocessedFrame` carries the output of preprocessing, `VisualConcepts` and `DraftReasoning` carry the output of the VLM, `CommonsenseResult` carries the ConceptNet evidence, and `SocialPrediction` carries the final prediction. Together these form a `Phase1Result`, which can be serialised to JSON with all of its evidence and module timings.

Algorithm 10 Phase I Pipeline

Input: Input source S, optional question Q, optional answer options O

Output: Phase I result: frame, visual concepts, draft, commonsense evidence, prediction, timings

1: $F \leftarrow \textsc{Preprocess}(S)$ ▷ Algorithm 1
2: $(V, D) \leftarrow \textsc{SmolVLM}(F, Q)$ ▷ Algorithm 3
3: $R \leftarrow \textsc{EnrichWithConceptNet}(V, D, Q)$ ▷ Algorithm 6
4: $P \leftarrow \textsc{ReasonSocialInteraction}(F, V, D, R, Q, O)$ ▷ Algorithm 9
5: **return** (F, V, D, R, P) with the time taken by each stage

3.5 BACKEND AND USER INTERFACE

The FastAPI backend loads the Phase I pipeline once and keeps the model in memory between requests. Each analysis runs as a background job. While it runs, the React frontend polls the job, so the user can see which module is working at any moment. Appendix B lists the endpoints.

The user can upload an image, take a photo with the live camera, or upload a video, in which case the video's key moments are analysed. A question can

optionally be added. The results appear as three prediction cards, one each for the activity, relationship and intention, showing the confidence value and the alternatives. Below the cards come the generated explanation, the ConceptNet evidence behind it, the visual concepts found by the VLM and the time taken. The whole result can be saved as a JSON report, and a history strip lets the user return to earlier analyses from the same session. This interface only demonstrates Phase I in action. It is not the explainability dashboard planned for Phase II.

3.6 PLANNED PHASE II EXTENSION

Most of what Phase II needs is already produced in Phase I: the normalised tensor of Equation 3.1, which visual attribution will use; access to the model through `SmolVLMEngine`; the structured prediction targets; the written rationale; and the ConceptNet evidence. Phase II will add:

- **Visual explanation path:** Grad-CAM [21] maps for each predicted label, and extraction of the concepts present in the most activated regions, giving a visual concept set $\mathcal{V}$.
- **Textual explanation path:** extraction of the concepts mentioned in the generated explanation, giving a textual concept set $\mathcal{T}$.
- **Cross-modal consistency:** the Jaccard overlap of the two sets,

$$\text{CMC} = \frac{|\mathcal{T} \cap \mathcal{V}|}{|\mathcal{T} \cup \mathcal{V}|} \tag{3.3}$$

- **Faithfulness evaluation:** the Insertion–Deletion metric [22] and the Pointing Game [23].
- **Explainability dashboard:** the input frame, the predicted interaction, the three labels and the consistency score in a single view.

CHAPTER 4

EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

This chapter describes the data and the computing environment used, the experiments conducted in Phase I, and an analysis of their results.

4.1 DATASET DESCRIPTION

TABLE 4.1: Dataset Description

Attribute	**Description**
Dataset	Visual Commonsense Reasoning (VCR) [1], Hugging Face release (Parquet files)
Source images	About 110,000 movie scenes from LSMDC [31] and MovieClips
Questions	About 290,000 multiple-choice questions
Options	4 candidate answers; 4 candidate rationales for the correct answer
Tasks	Q→A, QA→R and the joint metric Q→AR
Training set	VCR training split, 3 examples per question
Selection set	VCR validation questions, used only to choose the epoch
Test set	Separate held-out VCR validation questions
Knowledge source	ConceptNet 5.5 [14], local SQLite database

4.1.1 Visual Commonsense Reasoning (VCR) Dataset

The VCR dataset [1] contains about 290,000 multiple-choice questions over about 110,000 images taken from movie scenes, drawn from the LSMDC [31] and

MovieClips collections, as summarised in Table 4.1. Each question comes with four candidate answers and, for the correct answer, four candidate rationales. Objects and people in each image are detected and referred to in the text by tags, for example “Why is [person1] pointing at [person2]?”. The questions cover explanations, activities, mental states, roles, temporal inferences and hypothetical situations, which makes them a close match to the activity, relationship and intention targets of this project. The wrong options are produced by Adversarial Matching, so they are relevant to the scene and cannot be rejected on surface cues alone.

4.1.2 Data Splits Used

Since the labels of the official VCR test split are not public, all evaluation questions are drawn from the official validation split. Three disjoint sets are used. The **training set** contains questions from the VCR training split, each expanded into a Q→A, a QA→R and a free-form explanation example. The **selection set** of validation questions is used only to choose the epoch. The **test set** of further validation questions is used neither for training nor for choosing the epoch.

4.1.3 ConceptNet Knowledge Base

We use ConceptNet 5.5 [14] as our source of commonsense knowledge. A local SQLite database built from the ConceptNet assertions allows offline and reproducible lookups, with the public ConceptNet API and an on-disk cache as a fallback.

4.1.4 Social Interaction Taxonomy

The label space for the three prediction tasks is defined in `configs/taxonomy.yaml`. For every label, the file stores a short description, a few ConceptNet anchor concepts and a list of keywords. Activity labels include *Performing*, *Watching together* and *Photographing*. Relationship labels include *Performer or speaker and audience*, *Teammates or members of the same group*, *Doctor (or nurse) and patient* and `none_single`. Intention labels include *To entertain or perform for others*, *To socialise, relax and spend time together* and *To comfort, console or emotionally support someone*.

4.1.5 Test Fixtures and Demonstration Data

The automated tests use synthetic images, a deterministic stand-in for the VLM and a small ConceptNet graph, so they run without downloading the model. The qualitative demonstration uses real photographs, including the concert photograph of Section 4.4.6.

4.2 ECOSYSTEM

4.2.1 Hardware Configuration

Fine-tuning was carried out on a GPU in the department laboratory, and the demonstration and interactive experiments were run on a local machine. The hardware configuration used is:

- **Fine-tuning:** GPU workstation in the department laboratory (CUDA).
- **Inference and demonstration:** Apple Silicon MacBook, using the Metal Performance Shaders (MPS) backend of PyTorch.
- **Knowledge base storage:** ConceptNet is kept as a local SQLite database on the machine that runs inference, so the system does not need internet access while running.

4.2.2 Software Environment

The code is organised as a Python package, with a separate module for each of preprocessing, VLM inference, commonsense enrichment, reasoning, configuration, data structures and visualisation. The main tools and libraries we used are listed below.

- **Programming Language:** Python 3.12
- **Image/Video Processing:** OpenCV [27], NumPy and PIL
- **Vision-Language Model:** SmolVLM, run through Hugging Face Transformers [29]
- **Deep Learning Framework:** PyTorch [30]
- **Fine-Tuning:** PEFT (LoRA)
- **Natural-Language Processing:** spaCy, plus our own text utilities
- **Commonsense Knowledge:** ConceptNet, stored in SQLite with a local cache

- **Backend:** FastAPI, Uvicorn
- **Frontend:** React, Vite, JavaScript
- **Testing:** pytest
- **Configuration:** YAML

4.3 EXPERIMENTS CONDUCTED

4.3.1 Functional Validation Experiment

We ran the automated tests with pytest to check each module on its own and the pipeline as a whole. The preprocessing tests cover image loading, resizing, normalisation, low-light enhancement, blur detection and frame sampling. The ConceptNet tests cover term resolution, enrichment, label support and reading from the local SQLite database. A few further tests check text normalisation and taxonomy loading. The end-to-end tests run the complete pipeline and check that commonsense evidence resolves the prediction when the VLM is uncertain, that scenes with one person or with no people are handled correctly, and that multiple-choice questions are answered. The video tests check key-moment selection and summarisation.

4.3.2 LoRA Fine-Tuning Experiment

SmolVLM was fine-tuned for 20 epochs with the configuration of Table 3.2, and the adapter was evaluated on the selection set after every epoch to choose the checkpoint.

4.3.3 Quantitative Test Evaluation

The base model and the selected fine-tuned model were evaluated on the same held-out test questions on Q→A, QA→R and Q→AR.

4.3.4 Qualitative Comparison of Explanations

The free-form answers and reasons of the base and fine-tuned models were compared for test questions against the reference answers.

4.3.5 End-to-End Demonstration

The integrated pipeline was run through the web interface on real photographs to inspect the predictions, the evidence and the processing time.

4.4 PERFORMANCE ANALYSIS

4.4.1 Functional Validation Results

All 23 automated tests passed, as summarised in Table 4.2. These tests confirm that the module interfaces work together under controlled conditions: an input flows through preprocessing, VLM stand-in inference, ConceptNet enrichment and social reasoning to a structured prediction. Because the VLM is replaced by a deterministic stand-in, they validate the software, not the accuracy of the model; accuracy is analysed in the following sections.

TABLE 4.2: Automated Validation Results

Test Category	**Tests**	**Result**
Preprocessing	7	Passed
ConceptNet	3	Passed
Text and taxonomy	5	Passed
End-to-end Phase I pipeline	5	Passed
Video key-moment selection	3	Passed
Total	**23**	**23 Passed**

4.4.2 Training Convergence

The training loss is shown per optimiser step in Figure 4.1, together with its moving average over 20 steps. Averaged per epoch, it falls from 2.03 in the first epoch to 0.08 in epoch 7 and 0.013 in epoch 11, and it is 0.001 or less from epoch 15 onwards. The loss therefore converges well before the end of the run.

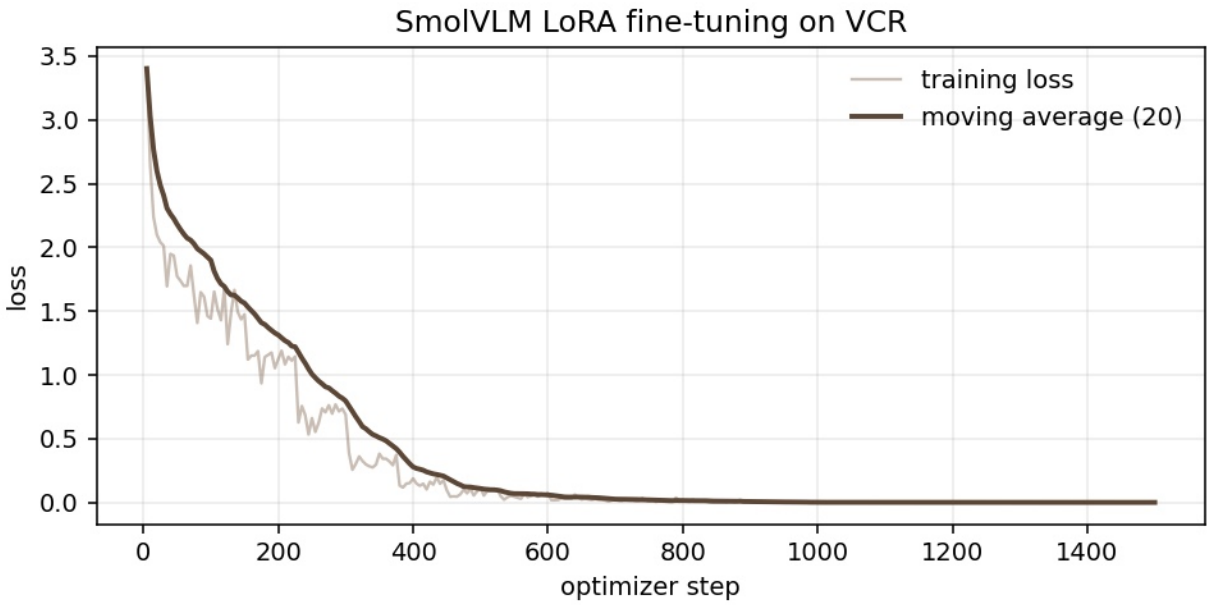


FIGURE 4.1: Training Loss of LoRA Fine-Tuning on VCR over 1,500 Optimiser Steps

4.4.3 Epoch Selection

The selection-set accuracy after every epoch is shown in Figure 4.2 and Table 4.3. Accuracy rises quickly over the first three epochs, with Q→AR increasing from 33% to 46%, and then fluctuates between 43% and 48% while the training loss keeps falling. From epoch 12 onwards the accuracy is identical in every epoch, because the training loss is close to zero and the adapters no longer change.

Epochs 9 and 11 share the highest Q→AR accuracy of 48%. Epoch 11 is selected because its Q→A accuracy is higher (63% against 62%), and its QA→R accuracy of 65% is the second highest of the run. The final model therefore uses the epoch-11 adapter rather than the adapter from the last epoch.

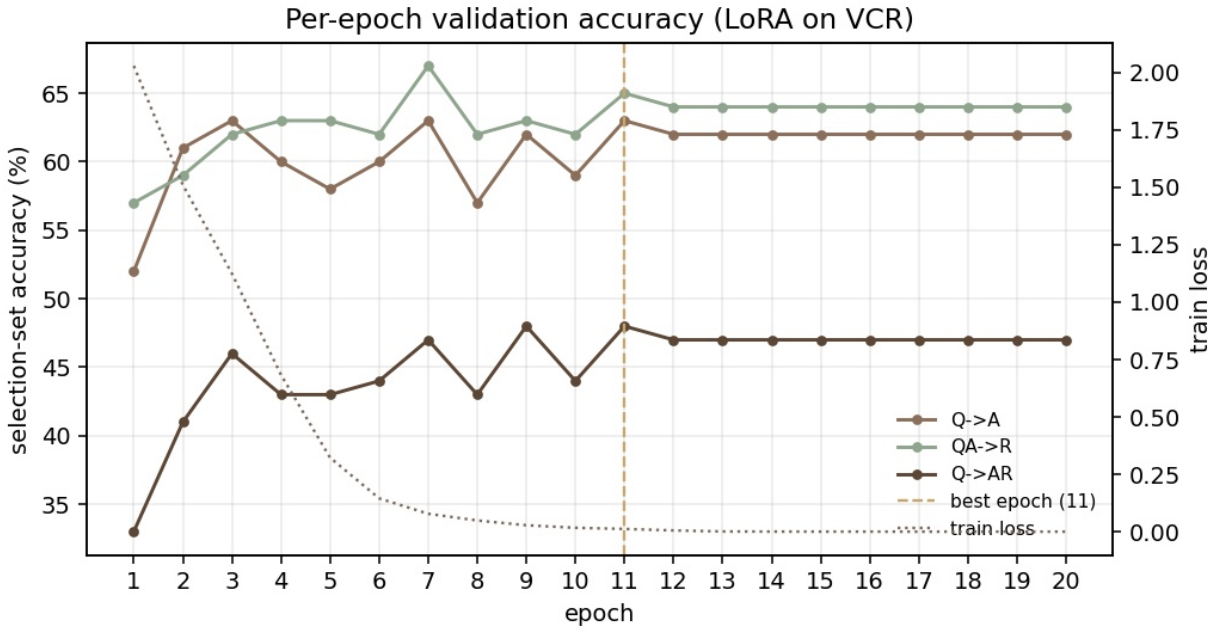


FIGURE 4.2: Selection-Set Accuracy and Training Loss after Each Epoch (Dashed Line: Selected Epoch 11)

TABLE 4.3: Per-Epoch Accuracy on the Selection Set

Epoch	Train Loss	Q→A (%)	QA→R (%)	Q→AR (%)
1	2.029	52	57	33
2	1.510	61	59	41
3	1.124	63	62	46
4	0.684	60	63	43
5	0.323	58	63	43
6	0.146	60	62	44
7	0.079	63	67	47
8	0.050	57	62	43
9	0.029	62	63	48
10	0.018	59	62	44
11	**0.013**	**63**	**65**	**48**
12–20	0.006 – 0.001	62	64	47

The separation between the two curves – training loss approaching zero while the selection accuracy stops improving after epoch 3 – shows that the later epochs mostly fit the training examples more closely without generalising better. Selecting the epoch on held-out data, rather than using the last checkpoint, guards against this.

4.4.4 Test Accuracy

The base model and the fine-tuned model (epoch 11) were evaluated on the same held-out validation questions. Table 4.4 and Figure 4.3 give the results. Fine-tuning improves all three tasks by 14 to 17 percentage points, and the Q→AR accuracy more than doubles, from 15.3% to 32.7%.

TABLE 4.4: Accuracy on Held-Out VCR Validation Questions

Model	**Q→A (%)**	**QA→R (%)**	**Q→AR (%)**
Chance level	25.0	25.0	6.3
SmolVLM-Instruct (base)	44.7	41.3	15.3
SmolVLM + LoRA (epoch 11)	**58.7**	**57.7**	**32.7**
Improvement (percentage points)	+14.0	+16.3	+17.3

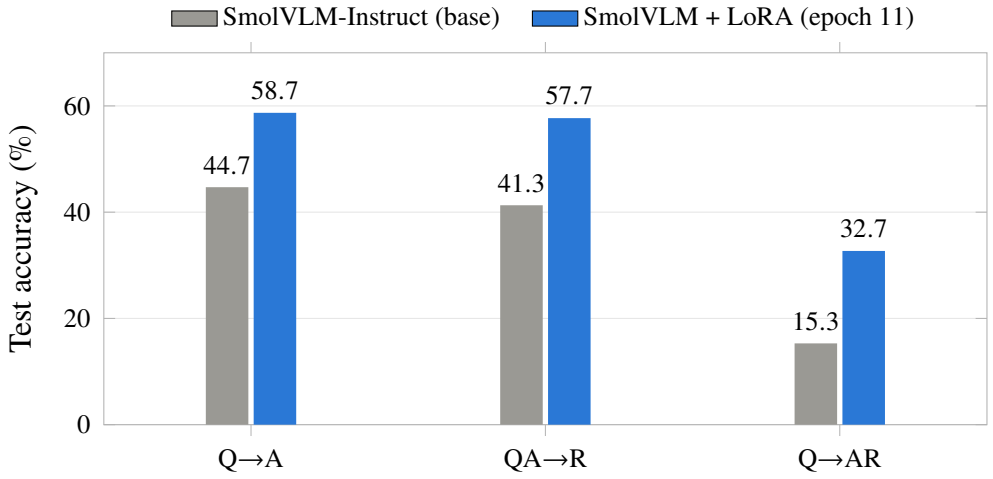


FIGURE 4.3: Test Accuracy of the Base and Fine-Tuned Models

4.4.4.1 Selection versus Test Accuracy

The test accuracy is lower than the selection-set accuracy of epoch 11 (63%, 65% and 48%). This is expected: the epoch was chosen because it scored well on the selection set, so its selection accuracy is optimistically biased. This is why the result is reported on a separate test set.

4.4.4.2 Answer and Rationale

If answer and rationale selection were independent, the expected Q→AR accuracy would be the product of the Q→A and QA→R accuracies: $0.587 \times 0.577 \approx 33.9\%$ for the fine-tuned model, close to the observed 32.7%, and $0.447 \times 0.413 \approx 18.5\%$ for the base model, above the observed 15.3%. Fine-tuning therefore improves both sub-tasks rather than only one, and the remaining errors are spread across both.

4.4.4.3 Comparison with Published Results

For reference, the R2C model reported 65.1%, 67.3% and 44.0% (Q→A, QA→R, Q→AR) on the full VCR test set, and human performance was 91.0%, 93.0% and 85.0% [1]. These figures are not directly comparable with ours, which are on a held-out subset of the validation split, and R2C was trained on the full training set with detected object features. The comparison shows that a 2.2-billion-parameter general-purpose VLM, adapted with LoRA on a small amount of VCR data, reaches about three quarters of R2C's joint accuracy while also being able to generate free-form explanations, and that a large gap to human performance remains.

4.4.5 Qualitative Comparison of Explanations

Table 4.5 compares the free-form explanations of the two models for four test questions. For the violin and quarrel questions, the fine-tuned model gives the correct answer together with a visual reason, whereas the base model only describes what a person is looking at. For the father–son question, the base model states the relationship correctly while the fine-tuned model reverses it, although it

gives the expected kind of reason (the age gap). For the dock question, neither model reaches the reference answer. The fine-tuned answers consistently follow the VCR pattern of an answer followed by its visual reason.

TABLE 4.5: Explanations of the Base and Fine-Tuned Models for Four Test Questions

What is person2 doing?	
Reference	He is playing a violin. He is holding a violin in one hand and applying the bow to it with the other.
Base	Looking at the ship.
Fine-tuned	He is playing a violin. He is holding a violin and a bow.
What is happening between person2 and person1?	
Reference	They are quarreling. person2 looks tense and stern, person1 looks surprised.
Base	Person2 is looking at person1.
Fine-tuned	person1 is telling person2 something she doesn't like. person2 is looking at person1 with a stern expression.
How do person7 and person1 know each other?	
Reference	person1 is person7's father. They came in together, and person1 is much older than person7.
Base	Person7 and person1 know each other. Person7 is the son of person1.
Fine-tuned	person7 is person1's father. There is a specific age gap between them, and they are going to embrace.
Why is person1 and person2 running on the dock?	
Reference	They are trying to board a boat before it leaves. They won tickets on the boat they are trying to reach.
Base	Person1 and person2 are running on the dock because they are trying to catch person5.
Fine-tuned	person1 and person2 are trying to escape. person1 and person2 are running and person1 is carrying person2 up some stairs.

The father–son example illustrates why Phase II is needed: an explanation can cite the right kind of evidence and still reach the wrong conclusion. Checking that the cited evidence corresponds to what the model attends to in the image, and that the textual and visual explanations agree, is the purpose of the cross-modal consistency and faithfulness evaluation.

4.4.6 End-to-End Demonstration

Figure 4.4 shows the web interface analysing a photograph of a singer performing to a crowd at a concert. The system predicts the activity *Performing* with 93% confidence, the relationship *Performer or speaker and audience* with 91% and the intention *To entertain or perform for others* with 97% (Table 4.6). In each case the next alternative receives 3% or less, so none of the predictions is ambiguous. The generated explanation relates the prediction to visible evidence: the singer's outstretched arm and microphone, her smile, and the hands of the crowd reaching towards her.

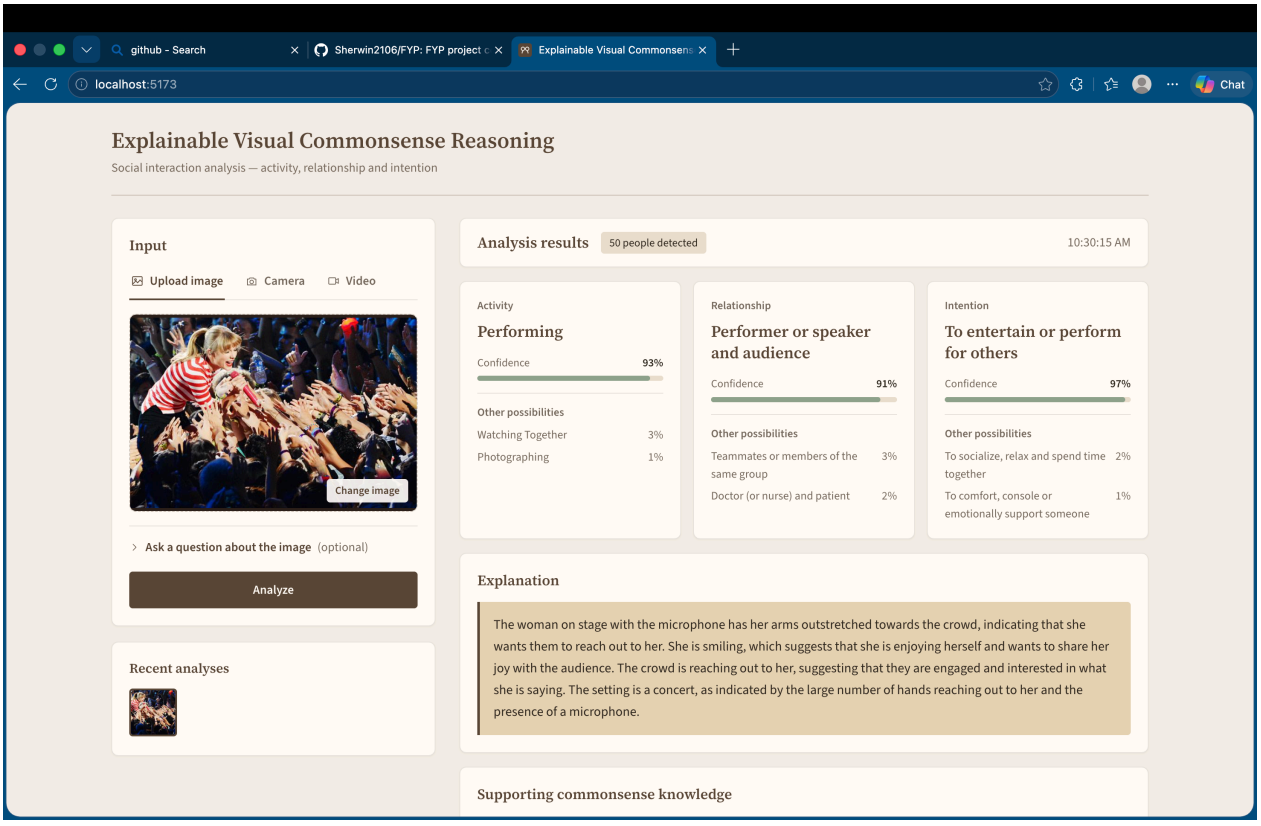


FIGURE 4.4: Web Interface: Predictions and Explanation for a Concert Photograph

The supporting evidence and the scene description are shown in Figure 4.5. ConceptNet links the observed concepts to the predicted labels: "concert" and

TABLE 4.6: Predictions for the Concert Photograph

Task	Prediction	Conf.	Next Alternatives
Activity	Performing	93%	Watching together (3%), Photographing (1%)
Relationship	Performer or speaker and audience	91%	Teammates (3%), Doctor and patient (2%)
Intention	To entertain or perform for others	97%	To socialise (2%), To comfort someone (1%)

"sing" are closely associated with performing, a performer is typically found on a stage, and a concert is a kind of show. The VLM identified the setting as a concert, the actions as reaching out and holding up, the expressions as happy and excited, and the objects as a water bottle, a microphone and hands. The number of people reached the system's upper limit of 50, since the crowd is larger than that.

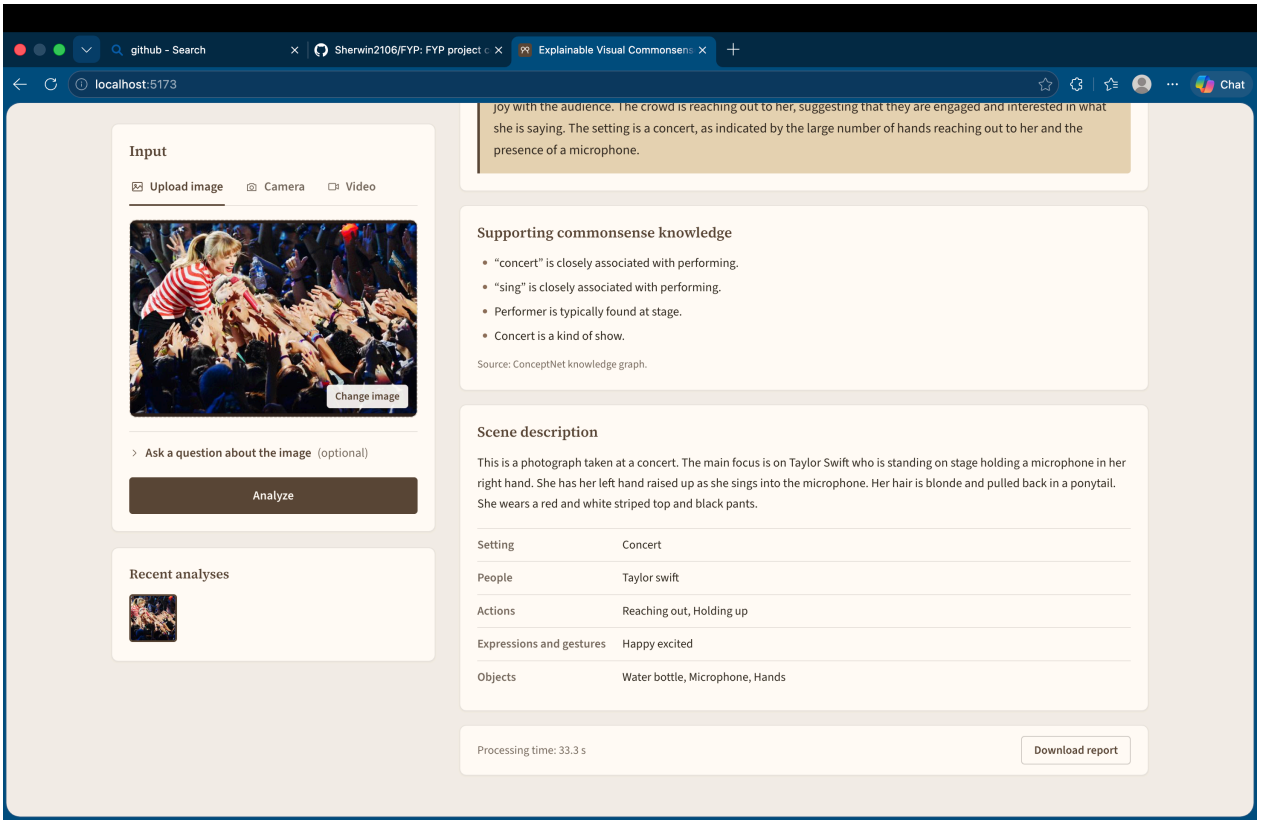


FIGURE 4.5: Web Interface: Supporting ConceptNet Knowledge and Scene Description

The complete analysis of the concert photograph took 33.3 seconds on an Apple Silicon MacBook using the MPS backend. Most of this time is spent in the VLM: for a scene with two or more people and no question, Algorithm 3 makes eleven generation calls, the explanation adds one more, and the option scoring of Algorithm 4 adds k forward passes for each of the three tasks. The preprocessing and ConceptNet stages operate on small arrays and a local database, and contribute comparatively little. The per-module timings are recorded in every `Phase1Result`.

4.4.7 Discussion

Compared with a conventional image-captioning or classification pipeline, the Phase I system reasons at three levels. At the *visual* level, it identifies the people, objects, actions, gestures and setting of the scene and describes them in a caption. At the *commonsense* level, it links these observations to ConceptNet relations and label-support paths. At the *decision* level, it fuses the VLM probabilities, the commonsense support and the draft match into a single prediction. The final prediction is therefore not a single sentence from the VLM: it retains the intermediate visual concepts, commonsense facts, component probabilities, alternatives and evidence that the explainability phase requires. Table 4.7 summarises the status of each Phase I module.

4.4.8 Limitations

- The quantitative evaluation measures the fine-tuned VLM on VCR questions. There is not yet a labelled benchmark for the activity,

TABLE 4.7: Phase I Implementation Status

Module	**Status**	**Implemented Result**
Input and Preprocessing	Completed	Image/video/live input, quality checks, enhancement, resizing, normalisation and frame sampling
Small Vision-Language Model	Completed	SmolVLM integration, visual concept extraction, draft reasoning, deterministic generation and option scoring
Fine-tuning (offline)	Completed	LoRA fine-tuning on VCR for 20 epochs; epoch 11 selected; Q→AR on held-out test questions raised from 15.3% to 32.7%
ConceptNet Commonsense Enrichment	Completed	Concept extraction, ConceptNet lookup, relevance filtering, label support, evidence and ambiguity resolution
Social Interaction Reasoning	Completed	Context fusion, activity/relationship/intention inference, log-linear fusion, confidence, alternatives and explanation generation
Explanation verification and dashboard	Phase II	Planned for the next phase

relationship and intention predictions of the complete pipeline, so the end-to-end accuracy of the fused prediction has been assessed only qualitatively.

- The fusion weights (w_{VLM}, w_{CN}, w_{draft}) and thresholds are set by hand; their effect has not yet been measured by an ablation study, and the calibration of the reported confidence values has not been measured.

- The analysis of one image takes about half a minute, which is acceptable for single images but too slow for continuous real-time video.

- The explanation is generated by the same VLM that makes the prediction, and its faithfulness has not yet been verified; this is the subject of Phase II.

CHAPTER 5

SOCIAL IMPACT AND SUSTAINABILITY

5.1 SOCIAL IMPACT

Systems that interpret social interactions in images are beginning to be used in everyday tools, and making such systems explainable and lightweight has several positive applications. For people who are blind or have low vision, a description that goes beyond listing objects – “a performer is entertaining an audience”, “two people appear to be arguing” – conveys the social meaning of a scene. Because the system runs on modest hardware, it can be deployed on personal devices without sending images to a remote server. Showing the cues behind an interpretation (gestures, expressions, posture and setting) can also support people who find non-verbal cues hard to read, and can be used in training material that teaches how such cues are interpreted.

A social or service robot has to tell whether people are busy, talking to each other or looking for help before it approaches them. If its reasoning can be inspected, developers can confirm that each decision was made for sound reasons. The same idea applies more widely. Every prediction our system makes comes with its evidence, the alternatives it considered and a confidence value, and unclear cases are flagged. Users can therefore see when the system is unsure and follow how it reached its answer. This encourages trust that matches how reliable the system actually is, rather than blind trust.

5.1.1 Ethical Considerations

Drawing conclusions about people's relationships and intentions from a picture is a sensitive task, and we see four main risks.

- **Privacy and identification:** Vision-language models can sometimes name well-known people. This happened in our own demonstration in Section 4.4.6, where the scene description named the performer. Knowing who someone is adds nothing to the social-interaction task, so the prompts and an output filter should stop the system from naming anyone, and faces can be blurred before any result is stored.

- **Surveillance misuse:** Automatic inference of relationships and intentions could be misused to watch people without their consent. Our system is meant for assistive and research use with users who have agreed to it. It processes images on the local machine and, by default, does not keep them.

- **Bias and stereotyping:** VCR comes from movie scenes, and pre-trained VLMs carry over patterns from their training data. Both can hold stereotypes, for instance about who looks like a "doctor" and who looks like a "patient", or about roles tied to gender and age. Commercial vision systems have already been shown to perform unevenly across demographic groups [32]. Because our system shows its evidence and alternatives, such biases are easier to notice, and we plan to audit its predictions across demographic groups.

- **Over-reliance:** A wrong guess about someone's intention can mislead, especially when the system sounds confident. We show the ambiguity flag

and the alternatives so that users read the output as one possible interpretation, not as a fact.

5.2 SUSTAINABILITY

Several of our design choices keep the computational and environmental cost of the system low [33]. The 2.2-billion-parameter SmolVLM is far smaller than the large multimodal models commonly used for visual reasoning, so inference runs on a laptop GPU, Apple-silicon MPS or a CPU instead of a data-centre accelerator. LoRA trains only the low-rank adapters while the vision encoder, the connector and the language-model weights remain frozen; the complete training run was 1,500 optimiser steps on a single laboratory GPU. The selected adapter is merged into the base weights at load time, so the fine-tuned model costs no more to run than the base model.

Sustainability is further improved by reuse and offline operation. Commonsense knowledge comes from an existing knowledge graph queried from a local database, and the label space is a configuration file, so new labels can be added without retraining the model. Because both ConceptNet and the model run locally, the system works without an internet connection. This reduces data transfer and removes any dependence on paid cloud services. For video, the frame sampler drops blurry frames and frames that repeat the previous scene, so the model only spends time on frames that add something new.

5.2.1 Relevance to the Sustainable Development Goals

Our project relates to three of the United Nations Sustainable Development Goals:

- **SDG 3 – Good Health and Well-being:** helps people with visual impairment understand scenes, and can help people learn to read social cues.
- **SDG 9 – Industry, Innovation and Infrastructure:** explainable AI that is affordable and runs on ordinary hardware.
- **SDG 10 – Reduced Inequalities:** the system needs less powerful hardware, so more people can use multimodal AI, and its visible evidence makes biased predictions easier to spot and fix.

CHAPTER 6

CONCLUSION AND FUTURE WORK

In conclusion, Phase I of the project has produced a complete, tested prediction pipeline for explainable visual commonsense reasoning about social interactions. The system accepts image, video and live-camera input, standardises it with OpenCV, performs visual understanding and language reasoning with SmolVLM, enriches the detected concepts with ConceptNet, and fuses the resulting evidence in a configurable social-interaction reasoning module.

SmolVLM was fine-tuned on the VCR dataset with LoRA for 20 epochs, and the adapter from epoch 11 was selected on a held-out selection set. On separate held-out test questions it improves accuracy over the base model from 44.7% to 58.7% for Q→A, from 41.3% to 57.7% for QA→R and from 15.3% to 32.7% for Q→AR.

The main output of Phase I is a structured prediction of the activity, relationship and intention, accompanied by confidence values, alternatives, evidence and a textual explanation. A reusable `Phase1Pipeline` ties all the modules together, the FastAPI–React interface shows the full flow working from start to finish, and all 23 automated tests pass. This meets the Phase I objectives (1 to 10 in Section 2.3), and Phase II can build on an implementation that has already been tested.

In Phase II, the focus moves from making predictions to checking the explanations behind them. Grad-CAM maps computed over the SmolVLM vision encoder will give a visual explanation path, and the concepts mentioned in the generated explanation will give a separate textual path. We will measure how well

the two concept sets agree using the Jaccard-based cross-modal consistency score (Equation 3.3). Faithfulness will be evaluated with the Insertion–Deletion metric and the Pointing Game, and the input frame, the predicted interaction, the three labels, the explanations and the consistency score will be presented in an explainability dashboard.

We also plan the following improvements:

- building a labelled evaluation set for the activity, relationship and intention predictions of the full pipeline, and running an ablation study on the fusion weights and on each source of evidence;
- measuring how well the confidence values are calibrated, and improving them;
- evaluating on a larger VCR test set and training on more VCR questions;
- batching the VLM prompts to speed up inference, with the aim of getting close to real-time video analysis;
- adding privacy safeguards so that the system does not name individuals, and auditing its predictions across demographic groups.

REFERENCES

1. R. Zellers, Y. Bisk, A. Farhadi, and Y. Choi, “From recognition to cognition: Visual commonsense reasoning,” in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2019.

2. K. Lin *et al.*, “TAB-VCR: Tags and attributes based VCR baselines,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

3. W. Yu *et al.*, “Heterogeneous graph learning for visual commonsense reasoning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

4. J. S. Park, C. Bhagavatula, R. Mottaghi, A. Farhadi, and Y. Choi, “VisualCOMET: Reasoning about the dynamic context of a still image,” in *European Conf. on Computer Vision (ECCV)*, 2020.

5. J. Lu, D. Batra, D. Parikh, and S. Lee, “ViLBERT: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

6. W. Su *et al.*, “VL-BERT: Pre-training of generic visual-linguistic representations,” in *Int. Conf. on Learning Representations (ICLR)*, 2020.

7. Y.-C. Chen *et al.*, “UNITER: UNiversal image-TExt representation learning,” in *European Conf. on Computer Vision (ECCV)*, 2020.

8. R. Zellers *et al.*, “MERLOT: Multimodal neural script knowledge models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

9. A. Radford *et al.*, “Learning transferable visual models from natural language supervision,” in *Int. Conf. on Machine Learning (ICML)*, 2021.

10. J. Li, D. Li, S. Savarese, and S. Hoi, “BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models,” in *Int. Conf. on Machine Learning (ICML)*, 2023.

11. H. Liu, C. Li, Q. Wu, and Y. J. Lee, “Visual instruction tuning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

12. A. Marafioti *et al.*, “SmolVLM: Redefining small and efficient multimodal models,” *arXiv preprint*, 2025.

13. X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer, “Sigmoid loss for language image pre-training,” in *Proc. IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, 2023.

14. R. Speer, J. Chin, and C. Havasi, “ConceptNet 5.5: An open multilingual graph of general knowledge,” in *Proc. AAAI Conf. on Artificial Intelligence*, 2017.

15. K. Marino, M. Rastegari, A. Farhadi, and R. Mottaghi, “OK-VQA: A visual question answering benchmark requiring external knowledge,” in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2019.

16. K. Marino, X. Chen, D. Parikh, A. Gupta, and M. Rohrbach, “KRISP: Integrating implicit and symbolic knowledge for open-domain knowledge-based VQA,” in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2021.

17. Q. Sun, B. Schiele, and M. Fritz, “A domain based approach to social relation recognition,” in *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2017.

18. J. Li, Y. Wong, Q. Zhao, and M. S. Kankanhalli, “Dual-glance model for deciphering social relationships,” in *Proc. IEEE Int. Conf. on Computer Vision (ICCV)*, 2017.

19. E. J. Hu *et al.*, “LoRA: Low-rank adaptation of large language models,” in *Int. Conf. on Learning Representations (ICLR)*, 2022.

20. T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “QLoRA: Efficient finetuning of quantized LLMs,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

21. R. R. Selvaraju *et al.*, “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in *Proc. IEEE Int. Conf. on Computer Vision (ICCV)*, 2017.

22. V. Petsiuk, A. Das, and K. Saenko, “RISE: Randomized input sampling for explanation of black-box models,” in *British Machine Vision Conf. (BMVC)*, 2018.

23. J. Zhang, S. A. Bargal, Z. Lin, J. Brandt, X. Shen, and S. Sclaroff, “Top-down neural attention by excitation backprop,” *Int. J. of Computer Vision*, vol. 126, pp. 1084–1102, 2018.

24. M. Kayser *et al.*, “e-ViL: A dataset and benchmark for natural language explanations in vision-language tasks,” in *Proc. IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, 2021.

25. Zhang *et al.*, “Cross-modal consistency in multimodal large language models,” 2024.

26. L. Parcalabescu and A. Frank, “Faithfulness of vision-language explanations,” 2024–2025.

27. G. Bradski, “The OpenCV library,” *Dr. Dobb’s Journal of Software Tools*, 2000.

28. K. Zuiderveld, “Contrast limited adaptive histogram equalization,” in *Graphics Gems IV*, P. S. Heckbert, Ed. Academic Press, 1994, pp. 474–485.

29. T. Wolf *et al.*, “Transformers: State-of-the-art natural language processing,” in *Proc. 2020 Conf. on Empirical Methods in Natural Language Processing: System Demonstrations*, 2020.

30. A. Paszke *et al.*, “PyTorch: An imperative style, high-performance deep learning library,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

31. A. Rohrbach *et al.*, “Movie description,” *Int. J. of Computer Vision*, vol. 123, pp. 94–120, 2017.

32. J. Buolamwini and T. Gebru, “Gender shades: Intersectional accuracy disparities in commercial gender classification,” in *Proc. Conf. on Fairness, Accountability and Transparency*, PMLR 81, 2018.

33. E. Strubell, A. Ganesh, and A. McCallum, “Energy and policy considerations for deep learning in NLP,” in *Proc. 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019.

34. V. S. G. *et al.*, “Explainable visual commonsense reasoning for social interactions – Review Report I,” Department of Computer Science and Engineering, Sri Sivasubramaniya Nadar College of Engineering, 2026.

APPENDIX A

CONFIGURATION PARAMETERS

Table A.1 lists the main parameters of the Phase I pipeline, set in `configs/default.yaml` unless stated otherwise.

TABLE A.1: Main Configuration Parameters of the Phase I Pipeline

Module	**Parameter**	**Value**
Preprocessing	Low-light trigger	brightness < 60 or contrast < 25
	Enhancement	CLAHE on L channel, gamma 0.8
	Blur threshold (Laplacian variance)	< 60
	Normalisation μ, σ	(0.5, 0.5, 0.5), (0.5, 0.5, 0.5)
	Video key moments (web interface)	up to 7
VLM	Model	`HuggingFaceTB/SmolVLM-Instruct`
	Decoding	greedy, repetition penalty
	Fine-tuned model	`configs/finetuned.yaml` (epoch-11 adapter, merged)
ConceptNet	Term salience	action 1.0, gesture 0.9, role 0.9, object 0.7, setting 0.6, caption 0.5
	Terms kept	top 20
	Facts kept	top 6 per term, top 40 overall
	Back-off penalty	salience $\times 0.5$
	Contradiction penalty	-0.5
Reasoning	Fusion weights	$w_{\text{VLM}} = 1.0$, $w_{\text{CN}} = 0.5$, $w_{\text{draft}} = 1.0$
	Commonsense temperature T	0.2
	Shortlist size	8 labels
	Ambiguity margin	0.15
	Maximum people count	50

APPENDIX B

API ENDPOINTS

Table B.1 lists the FastAPI endpoints implemented for the Phase I demonstration.

TABLE B.1: FastAPI Endpoints Implemented for the Phase I Demonstration

Endpoint	**Purpose**
`GET /api/health`	Reports backend, model and ConceptNet status.
`POST /api/analyze`	Accepts an image and an optional question and choices, and creates an analysis job.
`POST /api/analyze-video`	Accepts a video, selects key moments and analyses each of them.
`GET /api/jobs/{job_id}`	Returns the job status, the active module, the completed stages and the final result.